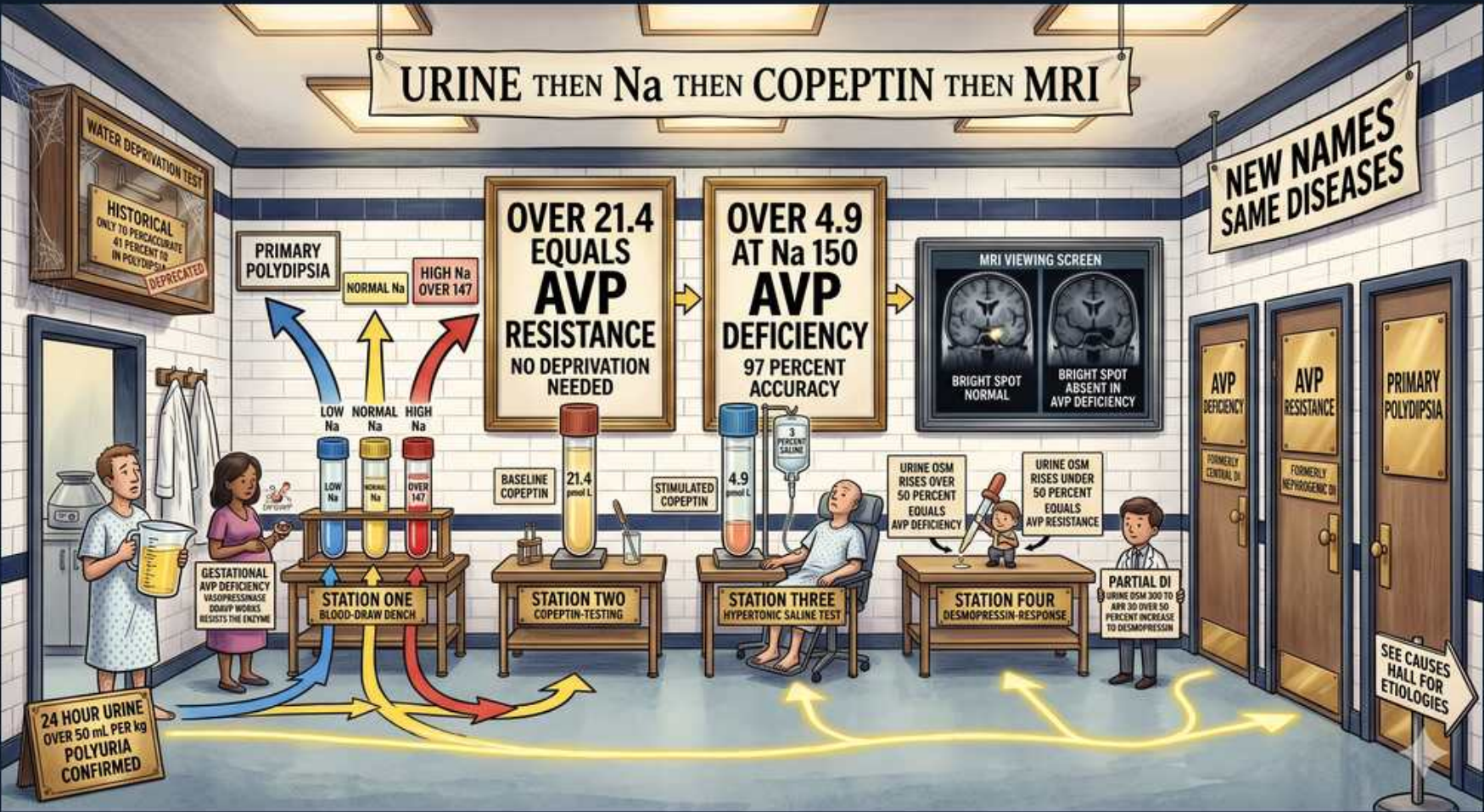


Polyuria workup — coceptin-based diagnostic algorithm



1. Sequence: urine, Na, copeptin, MRI

WHAT YOU SEE

Banner: urine then Na then copeptin then MRI

WHAT IT ENCODES

The modern polyuria workup proceeds from confirming polyuria, to serum sodium/osmolality, to copeptin-based testing, then MRI. Copeptin has largely replaced the classic water deprivation test for accuracy. The order localizes the lesion efficiently.

BOARD TRAP

Hypertonic-saline-stimulated copeptin outperforms the water deprivation test, which often misclassifies partial central DI and polydipsia.

2. Confirm true polyuria >3 L

WHAT YOU SEE

24 hour urine over 50 mL/kg, polyuria confirmed

WHAT IT ENCODES

True polyuria is urine output >3 L/day or >50 mL/kg/day on a 24-hour collection. This separates genuine polyuria from urinary frequency. Osmotic causes (glucose) are excluded at this step.

BOARD TRAP

Frequency without high 24h volume is not polyuria; and glucose/mannitol osmotic diuresis must be excluded before DI workup.

3. Primary polydipsia — low Na

WHAT YOU SEE

Low Na column, primary polydipsia branch

WHAT IT ENCODES

A low-normal serum sodium suggests primary polydipsia, where excess intake dilutes plasma and suppresses AVP. Baseline copeptin is low and rises appropriately with stimulation. Water excess, not water loss, drives symptoms.

BOARD TRAP

Low serum sodium favors polydipsia (water excess); DI typically presents with high-normal sodium from free water loss.

4. High Na (>147) branch

WHAT YOU SEE

High Na over 147, AVP deficiency direction

WHAT IT ENCODES

A high serum sodium with polyuria points toward AVP deficiency or resistance, since water loss concentrates plasma. This branch directs toward copeptin testing to separate central from nephrogenic. Implies inadequate water replacement.

BOARD TRAP

High-normal/high sodium with dilute urine = DI (deficiency or resistance); the sodium direction is the first fork.

5. Station 1 — baseline copeptin

WHAT YOU SEE

Station one copeptin testing, baseline

WHAT IT ENCODES

Baseline copeptin is drawn first. A baseline copeptin >21.4 pmol/L diagnoses nephrogenic DI (AVP resistance) without further deprivation. High AVP secretion with renal non-response defines this.

BOARD TRAP

Baseline copeptin >21.4 pmol/L = nephrogenic DI; no water deprivation needed at this cutoff.

6. Copeptin >21.4 = AVP resistance

WHAT YOU SEE

Over 21.4 equals AVP resistance, no deprivation needed

WHAT IT ENCODES

A markedly elevated baseline copeptin reflects intense endogenous AVP that the kidney ignores. This confirms nephrogenic DI and shortcuts the protocol. DDAVP will fail to concentrate urine.

BOARD TRAP

Very high copeptin means the pituitary is working — the lesion is renal (nephrogenic), so DDAVP is not the treatment.

7. Station 2/3 — stimulated copeptin

WHAT YOU SEE

Station two/three, stimulated copeptin, hypertonic saline

WHAT IT ENCODES

If baseline is intermediate, copeptin is stimulated with hypertonic saline (or arginine) to provoke AVP. A stimulated copeptin >4.9 pmol/L excludes central DI with ~97% accuracy. This separates partial central DI from primary polydipsia.

BOARD TRAP

Stimulated copeptin >4.9 pmol/L rules out central DI; below it confirms AVP deficiency — the key polydipsia-vs-central-DI discriminator.

8. Stimulated >4.9 at Na 150

WHAT YOU SEE

Over 4.9 at Na 150 equals AVP deficiency, 97 percent accuracy

WHAT IT ENCODES

During hypertonic saline infusion sodium is driven toward 150, and copeptin response is measured. A robust rise excludes central DI; a blunted rise confirms it. Requires close sodium monitoring during infusion.

BOARD TRAP

Hypertonic saline testing needs careful sodium monitoring; the copeptin response at target Na is what discriminates, not symptoms.

9. Station 4 — DDAVP/desmopressin trial

WHAT YOU SEE

Station four, desmopressin response

WHAT IT ENCODES

A desmopressin trial assesses renal responsiveness: urine concentrates in central DI but not nephrogenic DI. A urine osmolality rise >50% after DDAVP indicates central DI. A rise <50% indicates AVP resistance.

BOARD TRAP

DDAVP response >50% urine osmolality rise = central DI; <50% = nephrogenic DI — the functional confirmation step.

10. Urine Uosm rise >50% = central

WHAT YOU SEE

Urine Uosm rises over 50 percent equals central DI

WHAT IT ENCODES

Concentrating the urine after exogenous DDAVP confirms the kidney can respond, localizing the defect to AVP production (central). This complements copeptin testing. Treatment is then DDAVP replacement.

BOARD TRAP

A good DDAVP response localizes the lesion centrally — replace with DDAVP; a poor response means manage as nephrogenic DI.

11. MRI bright spot

WHAT YOU SEE

MRI screen, bright spot normal vs absent in AVP deficiency

WHAT IT ENCODES

MRI of the hypothalamic-pituitary region looks for the posterior pituitary bright spot, absent in central DI, plus stalk lesions or masses. Imaging follows biochemical confirmation. Identifies treatable structural causes.

BOARD TRAP

Absent bright spot supports central DI but is nonspecific with age — confirm biochemically with copeptin before attributing to a lesion.

12. Gestational AVP deficiency

WHAT YOU SEE

Gestational AVP deficiency note, water deprivation caution

WHAT IT ENCODES

Gestational DI arises from placental vasopressinase clearing native AVP and presents in the third trimester. It responds to DDAVP, which resists the enzyme. Resolves after delivery.

BOARD TRAP

In pregnancy, copeptin assays must account for vasopressinase; gestational DI responds to DDAVP, not native vasopressin.

13. New names same diseases

WHAT YOU SEE

New names same diseases sign, AVP deficiency/resistance/polydipsia doors

WHAT IT ENCODES

Nomenclature updated: central DI = AVP deficiency, nephrogenic DI = AVP resistance, with primary polydipsia unchanged. The renaming reduces confusion with diabetes mellitus. The underlying pathophysiology is identical.

BOARD TRAP

Know both terms: AVP deficiency = central DI, AVP resistance = nephrogenic DI — exams may use either nomenclature.
